01 SPEED

Faster simulation, broader experiments

MaoMi accelerates microscopy simulation while keeping anatomy, activity, optics, acquisition and measurement connected. Its RANA optical family adds spatial, vectorial and temporal detail. The independent RANA-X platform extends this approach to interchangeable material, biological and detector models. This report combines retained MaoMi evidence with new, bounded component and workflow experiments.

49.1×

matched optical-stage speedup

179.644 s → 3.662 s; 0.00667% full-volume difference.
Original MATLAB CPU versus MaoMi Standard CUDA: the same 512 × 512 × 60 tissue, pupil, scattering input and 4,225 propagation sites. Median of three serial calls on Ryzen 9 7950X / RTX 3080. [E1]

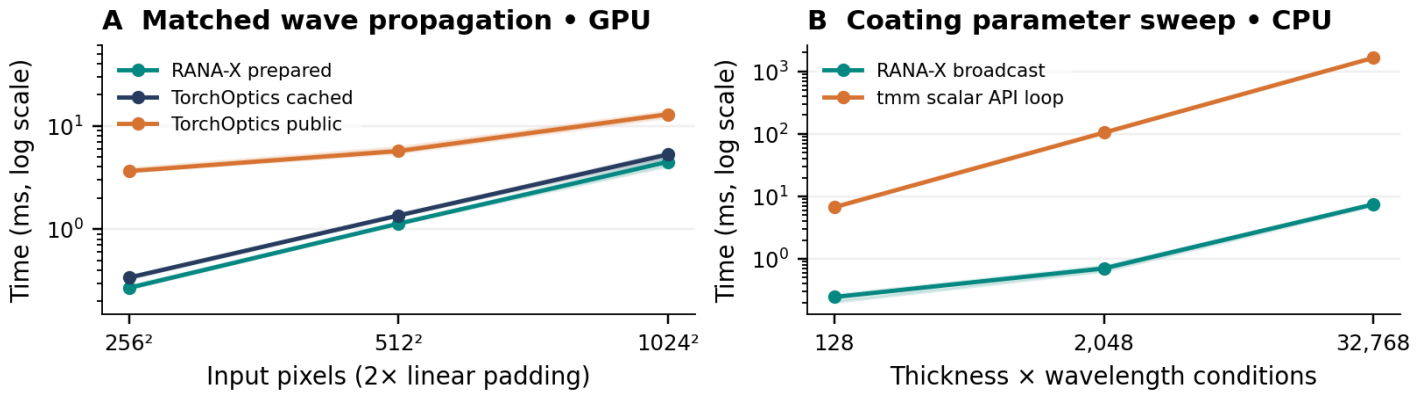


Figure 1. New measurements (26 October). Lines show medians; bands show the full repeat range. A: nine synchronized GPU repeats per route and size. B: five CPU repeats. All runs are serial; resident/prepared execution excludes startup and I/O. [E2]

Wave grid	RANA-X ms	TorchOptics ms cached / public	Film cases	Film ms RANA-X / tmm
256 × 256	**0.270**	0.339 / 3.646	128	**0.244** / 6.719
512 × 512	**1.131**	1.356 / 5.703	2,048	**0.702** / 104.709
1024 × 1024	**4.487**	5.282 / 12.946	32,768	**7.450** / 1651.708

Where the speed comes from

Batched propagation and reusable operators. At 1024^2, prepared RANA-X is 2.89× faster than TorchOptics public calls and 1.18× faster than its explicitly cached control. Most of the public-call gap is caching. Across all three sizes, complex-field relative L2 is at most 6.08×10^{-14}, using identical complex128 fields, 532 nm, 100 µm propagation and 2× linear padding. [E2, S2]

Broadcast physical parameter sweeps. A 128 × 256 thickness–wavelength coating sweep is 221.7× faster than a tmm scalar-API loop, with 5.72×10^{-16} reflectance relative L2. Both compute the same reflection observable; tmm also returns internal-layer quantities. This is a component comparison, not a ranking of complete simulators. [E2, S5]

E1 includes phase preparation, propagation, both tails and mask assembly; it excludes anatomy/input generation, static scanning, startup and I/O. The 49.1× value is retained historical evidence, not a new MATLAB run or an end-to-end movie claim.

02 CONNECTIONS BEYOND THE OPTICAL PATH

Replace the physics, keep the experiment

In RANA-X, packages exchange named physical quantities with units, axes and coordinate frames. A package can replace a model, introduce a quantity, or provide its own solver and workflow. Acquisition, state, noise, ground truth and export remain part of the same experiment.

Specimen / device
cells, film, potential
→
Physical response
kinetics, heat, fields
→
Acquisition
exposure, scan, time
→
Readout + truth
electrons, A, state

Connection	Executed path	Practical use
Neural activity / biology	Events → calcium kinetics → emission → camera	Swap exponential and biexponential responses; retain cell/activity truth.
Molecular response	Demo: bleaching / FRAP; fluorescence lifetime gates	Compare recovery and gate timing, including instrument response. [E5]
Optothermal materials	Temperature-dependent absorption ↔ heat balance	Replace conductive cooling with an external radiative model and solver. New run.
Coatings / biosensing	Layer thickness or analyte index → reflection	Sweep wavelength, angle and polarization; native film and plasmon examples. [E2, E5]
Infrared devices	Temperature + emissivity → photons → BP current	Black-phosphorus detector plug-in; amperes at 155 K and 1.7 V. [E5]
Camera electronics	QE → charge / noise → gain, clipping and ADC	Change sensor response or row exposure while retaining expected signal. [E5]
Electron microscopy	Electron probe → potential → elastic multislice	Entire recipe uses electron-wave and probability quantities; no RANA optical node. New run.

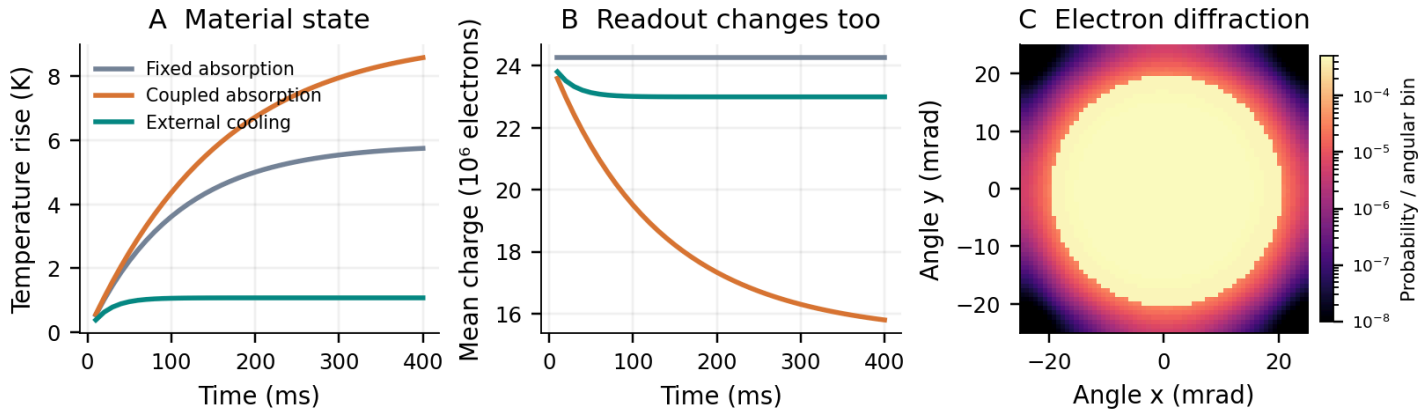


Figure 2. New cross-domain runs. A–B share illumination, acquisition and camera; only the material equations change. C: diffraction probability of one 200 keV electron probe after elastic multislice through prescribed potentials (log scale). These are illustrative models, not fitted device or specimen data. [E2]

A model swap changes the result. At 0.4 s, temperature rise is 5.75 K with fixed absorption, 8.58 K with thermal feedback, and 1.07 K with radiative cooling. All 40-step traces agree with independent heat-balance roots to 1.4×10^{-9} K. Replacing the external root solver with built-in iteration preserves this agreement.

The workflow survives the change. Resume reproduces all five thermal output arrays bitwise, including noisy readout. Electron probability is conserved to 1.56×10^{-15}. MaoMi supplies anatomy/activity, external PSFs and paired images; the broader model/solver examples belong to independent RANA-X.

03 RANA: PHYSICALLY RESOLVED IMAGE FORMATION

More useful detail, where it matters

RANA = Ray Aware Nonuniform Acquisition. A useful measurement depends on the specimen, local response, finite voxel, wavelength, polarization and actual exposure times. RANA carries these dependencies into image formation, so changing the experiment changes the predicted signal before detector noise is applied.

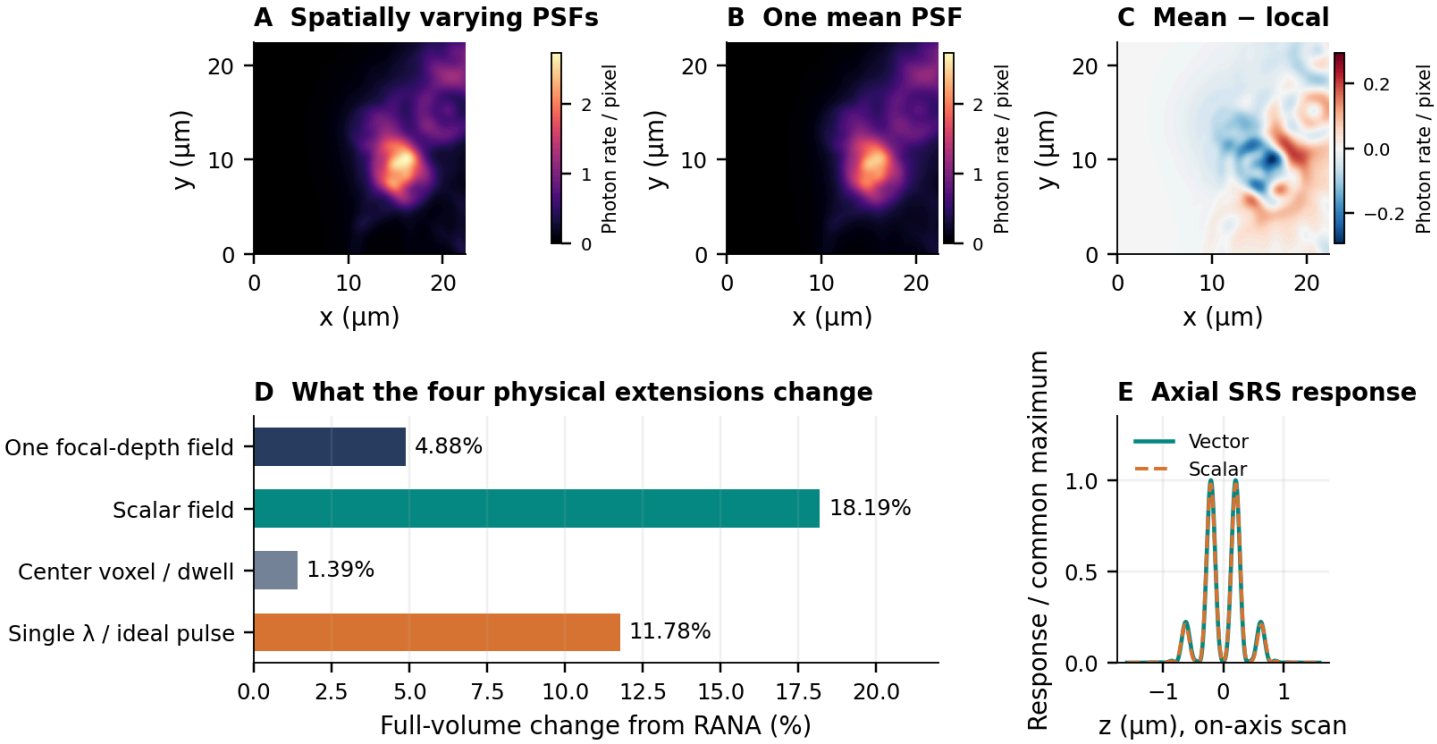


Figure 3. Computed examples, common scales. A–C: identical cell signal and 25 HCIPy-generated kernels; local rendering versus their arithmetic mean. The 22.4 µm crop is selected by signal alone. D: four separate MaoMi RANA ablations. E: vector/scalar on-axis 4Pi-SRS scans of two 80 nm spheres, NA 1.2. Plot normalization is shared within each comparison. [E3–E4]

Spatial, vector and spectral effects

Field-dependent optics. Replacing a 5 × 5 PSF field with one mean kernel changes the full 1024^2 photon image by **12.52%** at the declared aberration strength. Both routes use the same HCIPy PSFs: the gain is retaining field dependence, not outperforming HCIPy's pupil physics. [E4]

Depth and high NA. In the 256 × 256 × 16 MaoMi test (NA 1.1; specimen/immersion index 1.38/1.33; depth 50–62 µm), four simplifications together change the full volume by **30.31%**. In the separate SRS example, vector/scalar x–z scans differ by **14.41%** at NA 1.2, versus 0.106% at NA 0.2. [E3–E4]

Acquisition and numerical resolution

Time is part of the forward model. A retained flow example integrates moving photon fields at each row's exposure times. At 3.2 mm/s, reversing a 20 ms row sweep changes bead RMS height to **2.003× or 0.666×** the global-shutter value. Temporal refinement changes the signal by at most 0.01395%. [E5]

Check resolution separately from realism. Full RANA differs by **0.1030%** from a finer depth/angular/voxel/scan/spectral quadrature reference. A separate long-distance scalar test retains an old 120.82% ASM sampling discrepancy; automatic RS convolution reduces disagreement with its matched HCIPy reference to 8.74×10^{-11}%. [E3, E5]

Cost and scope. The RANA ablation example builds its PSF bank in 23.42 s on CPU; subsequent CPU/CUDA renders take 205.7/87.6 ms. It uses the standalone global-RBF renderer. Model differences quantify retained physics; numerical convergence does not establish experimental accuracy.

04 POSITION AMONG EXISTING SIMULATORS

An experiment platform around the optics

RANA's strongest case is the complete, replaceable forward experiment: connect a specimen or device, coupled equations, resolved optics, acquisition and measured outputs, then reuse the workflow. Established optical packages remain valuable both as execution backends and as independent references.

Reference	Established capability	Demonstrated MaoMi / RANA-X value
NAOMi [S1]	Neural anatomy, activity, tissue optics and calcium-imaging movies.	MaoMi keeps this foundation and accelerates a matched optical stage by 49.1×; adds Python workflows and optional RANA physics.
HCIPy [S3]	Wave optics, AO, coronagraphy and polarization.	Reuse or compare its optical outputs inside biological/material/readout experiments. Matched photon-image difference: 0.00291% in the circular-pupil test. [E4]
TorchOptics [S2]	Differentiable PyTorch Fourier optics; ASM / direct integration with automatic selection.	Prepared component execution is 1.18× faster than the cached 1024^2 control here; connect the field to stateful materials and measurements. [E2]
Chromatix [S4]	JAX-based differentiable GPU wave optics; high-NA/vector, sample and sensor models.	Our demonstrated emphasis is externally replaceable physical equations, solvers, non-optical quantities and resumable acquisition. No head-to-head speed or accuracy claim.
Pyxel / tmm [S5–S6]	Pyxel: detector-effect pipelines. tmm: multilayer optical calculations.	Our generic graph connects device and material models to other domains; the film sweep verifies one observable against tmm. No executed Pyxel bridge is claimed.

Three concrete reasons to choose this workflow

Generate controlled training pairs: vary cells, optics, activity and noise while retaining distinct latent and expected-measurement targets. **Study interacting models:** change temperature, absorption or an external detector without rebuilding the experiment. **Evaluate acquisition choices:** compare local PSFs, high-NA response, exposure timing and finite-pixel integration on the same specimen and readout.

What the evidence establishes. New runs use PyTorch 2.5.1 / TorchOptics 1.0.2, NumPy 2.2.5, SciPy 1.17.1 and tmm 0.2.0; CPU float64 and GPU complex128. Raw timing samples, recipes, arrays, source hashes and independent checks accompany this report. Overall superiority to Chromatix, full-vector multiple scattering, end-to-end differentiation, calibrated general device physics and real-data predictive accuracy are not established.

Primary references

[S1] Song et al. *Neural Anatomy and Optical Microscopy (NAOMi)*, J. Neurosci. Methods 358, 109173 (2021). doi:10.1016/j.jneumeth.2021.109173

[S2] TorchOptics, Propagation documentation.
[S3] HCIPy, official documentation and polarization tutorial.
[S4] Deb et al. *Chromatix*, Nature Methods 23, 1388–1398 (2026). doi:10.1038/s41592-026-03121-x; functional API.
[S5] Byrnes, *Multilayer optical calculations*. arXiv:1603.02720.
[S6] ESA Pyxel, detector pipeline documentation.

Experimental provenance

[E1] Retained MaoMi matched-site benchmark: *dense_4225/summary.json* and verification (October 2026).

[E2] New supplement: *propagation.json, films.json, thermal.json, electrons.json*; 26 October 2026. Generated by *run_experiments.py*.

[E3] MaoMi *docs/rana_comparison.json*: four ablations and refined numerical reference.

[E4] RANA-X matched HCIPy/TorchOptics, local PSF, SRS and infrared comparisons; 22 October 2026.

[E5] RANA-X application atlas and shutter-flow results: biology, sensors/materials, convergence and timing. Source paths and hashes: *evidence_manifest.json*.